

Non-Inertial Response of Correlations: From Scalar Bell Observables to an Extended Correlation Tensor

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Abstract

A standard Bell observable is a scalar correlation associated with a selected pair of local measurement directions. We formulate it as a projection of a complete two-particle correlation tensor and distinguish two angular sectors. The central result is a reversal of the sign multiplying the cosine law: for coincident calibrated settings, the Bell observable is positive for photons and negative for a fermionic singlet. This distinction provides an operational criterion for experimental identification. For photons, the positive sign follows from averaging two projection amplitudes over the complete non-inertial phase interval; the fermionic sign follows from the negative exchange holonomy of the phase-momentum sector. Mapping phase directions to linear-polarizer axes produces the corresponding double-angle dependence. The photon Stokes tensor has positive linear-polarization components and a negative circular-polarization component, whereas the fermionic singlet has an isotropic negative tensor. We introduce a motion-dependent extended tensor and a frequency-dependent non-inertial susceptibility. A phase-synchronous two-arm experiment combines optical modulation, rotation, and seeded multiaxial piezoelectric vibration. Independent motion measurements separate common and differential components and allow controlled variation from correlated to independent and oppositely driven motion; a single rigid platform is the simpler common-frame limit. Bell-setting and state-correlation vectors express the sign reversal as a scalar projection. The construction yields binary joint probabilities and recovers the Tsirelson bound with oppositely signed optimal CHSH combinations.

Keywords: Bell correlations; non-inertial reference frame; correlation tensor; entangled photons; fermionic singlet; non-inertial susceptibility; quantum tomography.

1 Introduction

In a standard Bell experiment [1, 2], the correlation between two binary outcomes is represented by a number $E(\mathbf{a}, \mathbf{b})$ for selected analyzer directions $\mathbf{a}$ and $\mathbf{b}$. Several such numbers form the CHSH combination. A single Bell observable, however, is not a complete description of the two-particle correlation structure; it is only one scalar projection of that structure.

This distinction becomes important when the local measurement systems rotate, accelerate, or undergo controlled non-inertial modulation. A variation of the recorded correlation may result from a change of the local measurement frames, from propagation and mode transformations, or from a response of the state and its intrinsic correlation structure. A single value of the CHSH parameter does not separate these contributions.

Our aim is to formulate Bell observables as projections of an extended correlation tensor and to define an experimentally reconstructible response of that tensor to non-inertial modulation. The ordinary two-qubit correlation matrix is well known in quantum-information theory [5]. The

additional object introduced here is its non-inertial susceptibility, including the amplitude and phase of the response to a periodic perturbation.

2 Photon Correlation on the Complete Phase Interval

In the non-inertial description, the complete state of a pair may include a common phase variable λ and a random metric structure. Let $A(a, \lambda)$ and $B(b, \lambda)$ denote local projection amplitudes. The physical axis of linear photon polarization is projective and has period π , but the phase variable of the correlation state completes the full cycle $0 \leq \lambda < 2\pi$. We therefore define

$$M_{AB}^{(\gamma)}(a, b) = \frac{1}{\pi} \int_0^{2\pi} A(a, \lambda) B(b, \lambda) d\lambda. \quad (1)$$

For the projection amplitudes

$$A(a, \lambda) = \cos(a - \lambda), \quad B(b, \lambda) = \cos(b - \lambda), \quad (2)$$

the identity $\cos x \cos y = \frac{1}{2}[\cos(x - y) + \cos(x + y)]$ gives

$$M_{AB}^{(\gamma)}(a, b) = +\cos(a - b). \quad (3)$$

Thus coincident phase directions are positively correlated, $M_{AB}^{(\gamma)}(a, a) = +1$. The integration interval belongs to the phase of the complete correlation state and must not be confused with the projective periodicity of a linear-polarizer axis. On the equator of the Poincare sphere, $a = 2\theta_A$ and $b = 2\theta_B$, and hence

$$E_\gamma(\theta_A, \theta_B) = +\cos 2(\theta_A - \theta_B). \quad (4)$$

This positive photon sector is the baseline used below. It is distinct from the negative correlation of a fermionic spin singlet.

3 Exchange Holonomy and the Fermionic Sign

The complete non-inertial state is defined in an extended phase space $\Gamma_{\text{ni}} = (\lambda, p_\lambda)$. For two indistinguishable particles, an exchange is represented by a closed path C_{ex} in the reduced two-particle configuration space. Such a space is multiply connected and its path integral admits the two one-dimensional exchange factors $+1$ and -1 [4]. We write the corresponding holonomy as

$$\mathcal{H}_\varepsilon(C_{\text{ex}}) = \exp\left(i \oint_{C_{\text{ex}}} \mathcal{A}\right) = \varepsilon, \quad \varepsilon = \pm 1, \quad (5)$$

where $\mathcal{A}$ is the effective connection of the transported extended state. The symmetric photon sector carries $\varepsilon_\gamma = +1$, whereas the antisymmetric fermionic sector carries $\varepsilon_f = -1$. This statement does not attempt to rederive the relativistic spin-statistics theorem; it identifies the exchange character retained by the complete non-inertial phase-momentum sector.

Transport of the second local projection around the exchange loop gives

$$A_\varepsilon(a, \lambda) = \cos(a - \lambda), \quad B_\varepsilon(b, \lambda) = \varepsilon \cos(b - \lambda). \quad (6)$$

Consequently,

$$M_{AB}^{(\varepsilon)}(a, b) = \frac{1}{\pi} \int_0^{2\pi} A_\varepsilon(a, \lambda) B_\varepsilon(b, \lambda) d\lambda = \varepsilon \cos(a - b). \quad (7)$$

Thus the positive photon and negative fermionic cosine laws follow from one phase average and the two exchange holonomies:

$$M_{AB}^{(\gamma)} = +\cos(a - b), \quad M_{AB}^{(f)} = -\cos(a - b). \quad (8)$$

The standard Bell states used later are representations of these two tensor sectors, rather than the origin of the signs in Eq. (8).

4 Bell-Setting and Correlation Vectors

The signed phase average above has a simple two-dimensional representation. Let

$$\Delta = a - b \quad (9)$$

be the relative phase direction of the two analyzers. Introduce two quadratures in arm A and one reference quadrature in arm B ,

$$A_c(a, \lambda) = \cos(a - \lambda), \quad A_s(a, \lambda) = \sin(a - \lambda), \quad B_c(b, \lambda) = \cos(b - \lambda). \quad (10)$$

Their normalized overlaps over the complete photon phase interval are

$$\begin{aligned} \frac{1}{\pi} \int_0^{2\pi} A_c(a, \lambda) B_c(b, \lambda) d\lambda &= \cos \Delta, \\ \frac{1}{\pi} \int_0^{2\pi} A_s(a, \lambda) B_c(b, \lambda) d\lambda &= \sin \Delta. \end{aligned} \quad (11)$$

The factor $1/\pi$ is the normalization weight of the correlation overlap; it is not a separately normalized probability density for λ . Equation (11) motivates the Bell-setting vector

$$\mathbf{b}_{AB}(\Delta) = \begin{pmatrix} \cos \Delta \\ \sin \Delta \end{pmatrix} = \frac{1}{\pi} \int_0^{2\pi} \begin{pmatrix} A_c(a, \lambda) B_c(b, \lambda) \\ A_s(a, \lambda) B_c(b, \lambda) \end{pmatrix} d\lambda. \quad (12)$$

This vector is not a spatial polarization or spin direction. It is an auxiliary vector of two angular functions fixed only by the relative analyzer setting.

Equivalently, with the local phase vectors

$$\mathbf{n}_A(a) = \begin{pmatrix} \cos a \\ \sin a \end{pmatrix}, \quad \mathbf{n}_B(b) = \begin{pmatrix} \cos b \\ \sin b \end{pmatrix}, \quad (13)$$

one may write

$$\mathbf{b}_{AB} = \begin{pmatrix} \mathbf{n}_A \cdot \mathbf{n}_B \\ \mathbf{e}_z \cdot (\mathbf{n}_B \times \mathbf{n}_A) \end{pmatrix} = \begin{pmatrix} \cos(a - b) \\ \sin(a - b) \end{pmatrix}. \quad (14)$$

Thus its cosine component is a scalar product and its sine component is the oriented cross product in the analyzer plane.

The state of the pair is described independently by the correlation vector

$$\mathbf{c} = V \begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix}, \quad 0 \leq V \leq 1, \quad (15)$$

where $V = |\mathbf{c}|$ is the correlation visibility and φ is the phase orientation of the correlation pattern. In the ideal, fully correlated case $V = 1$; values $0 \leq V < 1$ describe reduced visibility caused by imperfect state preparation, decoherence, background counts, or other experimental losses of correlation contrast. The scalar Bell observable is then

$$E(\Delta) = \mathbf{c} \cdot \mathbf{b}_{AB}(\Delta) = V \cos(\Delta - \varphi). \quad (16)$$

This definition is not circular: (V, φ) specify the prepared state, Δ specifies the apparatus, and Eq. (16) predicts the measured scalar. Estimating $\mathbf{c}$ from measured values of E is the inverse experimental problem.

For the two ideal exchange sectors, Eq. (7) corresponds to

$$\mathbf{c}_\varepsilon = \begin{pmatrix} \varepsilon \\ 0 \end{pmatrix}, \quad E_\varepsilon(\Delta) = \mathbf{c}_\varepsilon \cdot \mathbf{b}_{AB}(\Delta) = \varepsilon \cos \Delta. \quad (17)$$

For linear photon polarization,

$$\Delta_\gamma = 2(\theta_A - \theta_B), \quad \mathbf{c}_\gamma = \begin{pmatrix} +1 \\ 0 \end{pmatrix}, \quad (18)$$

and therefore

$$E_\gamma(\theta_A, \theta_B) = \mathbf{c}_\gamma \cdot \mathbf{b}_{AB}(\Delta_\gamma) = +\cos 2(\theta_A - \theta_B), \quad E_\gamma(0) = +1 > 0. \quad (19)$$

For coplanar fermionic spin analyzers,

$$\Delta_f = \alpha_A - \alpha_B, \quad \mathbf{c}_f = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad (20)$$

and hence

$$E_f(\alpha_A, \alpha_B) = \mathbf{c}_f \cdot \mathbf{b}_{AB}(\Delta_f) = -\cos(\alpha_A - \alpha_B), \quad E_f(0) = -1 < 0. \quad (21)$$

Equations (19) and (21) are two vector realizations of the same Bell observable, not two different definitions of it. The sign is meaningful only after the output-channel labels and the coincident settings have been calibrated consistently.

5 Binary Bell Probabilities and the Tsirelson Bound

Let the local outcomes be $A, B = \pm 1$. For fixed settings,

$$E(\mathbf{a}, \mathbf{b}) = \sum_{A, B = \pm 1} AB P(A, B \mid \mathbf{a}, \mathbf{b}). \quad (22)$$

For two-level systems, the local observables are

$$\hat{A}(\mathbf{a}) = a_i \hat{\sigma}_i, \quad \hat{B}(\mathbf{b}) = b_j \hat{\sigma}_j, \quad (23)$$

and hence

$$E(\mathbf{a}, \mathbf{b}) = \text{Tr} \left[\hat{\rho} \hat{A}(\mathbf{a}) \otimes \hat{B}(\mathbf{b}) \right]. \quad (24)$$

The CHSH parameter is

$$S = E(\mathbf{a}, \mathbf{b}) + E(\mathbf{a}, \mathbf{b}') + E(\mathbf{a}', \mathbf{b}) - E(\mathbf{a}', \mathbf{b}'). \quad (25)$$

It combines four scalar projections but does not reconstruct the complete tensor structure.

For unbiased local marginals, the signed non-inertial correlation $E_\varepsilon(a, b) = \varepsilon \cos(a - b)$ determines the binary joint probabilities

$$\begin{aligned} P_{++} &= P_{--} = \frac{1}{4} [1 + \varepsilon \cos(a - b)], \\ P_{+-} &= P_{-+} = \frac{1}{4} [1 - \varepsilon \cos(a - b)]. \end{aligned} \quad (26)$$

They are nonnegative, normalized, and reproduce the Bell observable,

$$\sum_{r, s = \pm 1} rs P_{rs}(a, b) = \varepsilon \cos(a - b). \quad (27)$$

Equation (27) supplies the explicit bridge from the continuous projection amplitudes to binary coincidence outcomes.

The same construction directly recovers the Tsirelson bound [3]. Introduce the unit phase vector $\mathbf{n}(a) = (\cos a, \sin a)$, so that $E_\varepsilon(a, b) = \varepsilon \mathbf{n}(a) \cdot \mathbf{n}(b)$. The CHSH combination becomes

$$S_\varepsilon = \varepsilon \{ \mathbf{n}(a) \cdot [\mathbf{n}(b) + \mathbf{n}(b')] + \mathbf{n}(a') \cdot [\mathbf{n}(b) - \mathbf{n}(b')] \}. \quad (28)$$

Maximization over a and a' gives

$$|S_\varepsilon| \leq |\mathbf{n}(b) + \mathbf{n}(b')| + |\mathbf{n}(b) - \mathbf{n}(b')|. \quad (29)$$

If the angle between $\mathbf{n}(b)$ and $\mathbf{n}(b')$ is δ , then

$$|S_\varepsilon| \leq 2 \left(\left| \cos \frac{\delta}{2} \right| + \left| \sin \frac{\delta}{2} \right| \right) \leq 2\sqrt{2}. \quad (30)$$

For $a = 0$, $a' = \pi/2$, $b = \pi/4$, and $b' = -\pi/4$, the two sectors give

$$S_\gamma = +2\sqrt{2}, \quad S_f = -2\sqrt{2}, \quad |S_\gamma| = |S_f| = 2\sqrt{2}. \quad (31)$$

Thus the exchange sign reverses the oriented CHSH combination without changing its maximal absolute magnitude. For physical photon polarizers the phase angles are doubled, giving the familiar settings 0° , 45° , 22.5° , and -22.5° .

6 Bell Observable as a Scalar Projection

Define the two-qubit correlation matrix

$$C_{ij} = \text{Tr}[\hat{\rho}(\hat{\sigma}_i \otimes \hat{\sigma}_j)] = \langle \hat{\sigma}_i \otimes \hat{\sigma}_j \rangle. \quad (32)$$

Then

$$E(\mathbf{a}, \mathbf{b}) = a_i C_{ij} b_j = \mathbf{a}^\top C \mathbf{b}. \quad (33)$$

The variation of the measured scalar separates three contributions:

$$\delta E = (\delta \mathbf{a})^\top C \mathbf{b} + \mathbf{a}^\top (\delta C) \mathbf{b} + \mathbf{a}^\top C (\delta \mathbf{b}). \quad (34)$$

The first and third terms describe changes of the local directions; the middle term describes a change of the correlation matrix itself. In orthonormal local bases, its elements are obtained from

$$C_{ij} = E(\mathbf{e}_i, \mathbf{e}_j). \quad (35)$$

6.1 Complete extended tensor

Set $\hat{\sigma}_0 = I$ and define

$$\mathcal{T}_{\mu\nu} = \text{Tr}[\hat{\rho}(\hat{\sigma}_\mu \otimes \hat{\sigma}_\nu)], \quad \mu, \nu = 0, 1, 2, 3. \quad (36)$$

It has the block form

$$\mathcal{T} = \begin{pmatrix} 1 & \mathbf{v}^\top \\ \mathbf{u} & C \end{pmatrix}, \quad u_i = \langle \hat{\sigma}_i \otimes I \rangle, \quad v_j = \langle I \otimes \hat{\sigma}_j \rangle. \quad (37)$$

The vectors $\mathbf{u}$ and $\mathbf{v}$ are the local polarizations, while C is the joint correlation block. The complete joint probability is

$$P(A, B | \mathbf{a}, \mathbf{b}) = \frac{1}{4} [1 + A \mathbf{a} \cdot \mathbf{u} + B \mathbf{b} \cdot \mathbf{v} + A B a_i C_{ij} b_j]. \quad (38)$$

Thus, a Bell observable probes only the correlation block, whereas the single counts also determine the local blocks of the full tensor.

7 Extended Non-Inertial Correlation Tensor

Let ξ^α denote controlled motion parameters, including orientation, angular velocity, acceleration, jerk, modulation phase, or stochastic variables. We introduce

$$\mathcal{C}_{ij}(\xi) = \text{Tr} \left[\hat{\rho}(\xi) \hat{\Sigma}_i^{(A)}(\xi) \otimes \hat{\Sigma}_j^{(B)}(\xi) \right], \quad (39)$$

and the corresponding scalar projection

$$E_{\text{ni}}(\mathbf{a}, \mathbf{b}; \xi) = a_i \mathcal{C}_{ij}(\xi) b_j. \quad (40)$$

The complete tensor is

$$\mathcal{T}(\xi) = \begin{pmatrix} 1 & \mathbf{v}^\top(\xi) \\ \mathbf{u}(\xi) & \mathcal{C}(\xi) \end{pmatrix}. \quad (41)$$

To separate motion of the measurement bases from an intrinsic response, write

$$\mathcal{C}_{\text{lab}}(\xi) = R_A(\xi) \mathcal{C}_{\text{int}}(\xi) R_B^\top(\xi). \quad (42)$$

After compensation of the calibrated frame rotations, the residual tensor is

$$\Delta \mathcal{C}_{\text{res}}(\xi) = R_A^{-1} \mathcal{C}_{\text{lab}} R_B^{-\top} - C^{(0)}. \quad (43)$$

A nonzero residual is not by itself evidence for new physics: decoherence, channel imperfections, drift, and standard relativistic transformations must first be included in the baseline prediction.

8 Photon Polarization

For photons, use normalized Stokes operators,

$$C_{\mu\nu}^{(\gamma)} = \langle \hat{S}_\mu^{(A)} \otimes \hat{S}_\nu^{(B)} \rangle, \quad \mu, \nu = 1, 2, 3. \quad (44)$$

With $\hat{S}_0 = I$, the complete two-photon Stokes tensor has indices $\mu, \nu = 0, 1, 2, 3$. The positive linear-polarization correlation is realized by

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|H\rangle_A |H\rangle_B + |V\rangle_A |V\rangle_B). \quad (45)$$

In the Stokes-axis order ($H/V, D/A, R/L$),

$$C^{(\gamma,0)} = \text{diag}(1, 1, -1), \quad \mathcal{T}^{(\gamma,0)} = \text{diag}(1, 1, 1, -1). \quad (46)$$

The vanishing local marginals express the absence of polarization of either photon separately. The two positive entries describe positive correlations of linear polarization; the circular component has the sign fixed by the chosen Bell state and Stokes-axis convention. A linear analyzer at angle θ corresponds to

$$\mathbf{n}(\theta) = (\cos 2\theta, \sin 2\theta, 0), \quad (47)$$

and therefore

$$E_0^{(\gamma)}(\theta_A, \theta_B) = \mathbf{n}^\top(\theta_A) C^{(\gamma,0)} \mathbf{n}(\theta_B) = +\cos 2(\theta_A - \theta_B). \quad (48)$$

The double angle expresses the equivalence of the physical axes θ and $\theta + \pi$; it does not reduce the full integration interval of the phase variable λ . The complete 3×3 block requires linear, diagonal, and circular polarization projections. A fit to a single cosine law does not replace tensor reconstruction.

9 Fermionic Singlet

For

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \quad (49)$$

one has

$$C_{ij}^{(f,0)} = -\delta_{ij}, \quad \mathcal{T}^{(f,0)} = \text{diag}(1, -1, -1, -1). \quad (50)$$

For coplanar spin analyzers,

$$E_0^{(f)}(\alpha_A, \alpha_B) = -\mathbf{a} \cdot \mathbf{b} = -\cos(\alpha_A - \alpha_B). \quad (51)$$

The single angle for spin, compared with the double angle for linear photon polarization, reflects the difference between an oriented spin direction and an unoriented polarization axis.

In a non-inertial setting,

$$\mathcal{T}^{(f)}(\boldsymbol{\xi}) = \begin{pmatrix} 1 & \mathbf{v}^\top(\boldsymbol{\xi}) \\ \mathbf{u}(\boldsymbol{\xi}) & -I + \Delta C^{(f)}(\boldsymbol{\xi}) \end{pmatrix}. \quad (52)$$

For relativistic massive particles, the spin transformation is momentum dependent. At fixed momenta,

$$C_{\text{lab}}^{(f)} = R_W^{(A)}(\mathbf{p}_A) C_{\text{int}}^{(f)} R_W^{(B)\top}(\mathbf{p}_B), \quad (53)$$

where R_W denotes the Wigner rotation. If momenta are unresolved, the observed tensor is averaged over the joint distribution,

$$\overline{C}_{ij}^{(f)} = \int d^3 p_A d^3 p_B f(\mathbf{p}_A, \mathbf{p}_B) C_{ij}^{(f)}(\mathbf{p}_A, \mathbf{p}_B). \quad (54)$$

Wigner rotations and spin-momentum averaging belong to the standard relativistic background and must be compensated before a residual non-inertial response is inferred [6, 7, 8].

10 Non-Inertial Susceptibility

Near the unperturbed state,

$$\mathcal{C}_{ij}(\boldsymbol{\xi}) = C_{ij}^{(0)} + \chi_{ij\alpha} \xi^\alpha + \frac{1}{2} \chi_{ij\alpha\beta}^{(2)} \xi^\alpha \xi^\beta + \dots, \quad (55)$$

with

$$\chi_{ij\alpha} = \left. \frac{\partial \mathcal{C}_{ij}}{\partial \xi^\alpha} \right|_{\boldsymbol{\xi}=0}. \quad (56)$$

For the full tensor,

$$X_{\mu\nu\alpha} = \left. \frac{\partial \mathcal{T}_{\mu\nu}}{\partial \xi^\alpha} \right|_{\boldsymbol{\xi}=0}, \quad (57)$$

where $X_{i0\alpha}$ and $X_{0j\alpha}$ describe induced local polarization and $X_{ij\alpha} = \chi_{ij\alpha}$ describes the correlation response.

For harmonic modulation, $\xi^\alpha(t) = \xi_0^\alpha \cos \Omega t$, a delayed response is described by a complex susceptibility,

$$\delta \mathcal{C}_{ij}(\Omega) = \chi_{ij\alpha}(\Omega) \xi^\alpha(\Omega). \quad (58)$$

Its real and imaginary parts represent in-phase and quadrature responses. The observable scalar susceptibility is

$$\chi_E^{(\alpha)}(\mathbf{a}, \mathbf{b}; \Omega) = a_i \chi_{ij\alpha}(\Omega) b_j. \quad (59)$$

11 Phenomenological Response Structure

The fermionic and photon baselines must be expanded separately. For the isotropic fermionic singlet, a minimal residual correction is

$$\Delta C_{ij}^{(f),\text{res}} = -\eta_f \delta_{ij} + q_f (m_i m_j - l_i l_j) + \kappa_f \epsilon_{ijk} n_k, \quad (60)$$

where η_f changes the overall spin anticorrelation, q_f describes a symmetric anisotropy, and κ_f is an antisymmetric rotational response.

For photons propagating parallel to the rotation axis, $\mathbf{n} \parallel \mathbf{k}$, choose transverse Stokes basis vectors $\mathbf{e}_1 = \mathbf{m}$ and $\mathbf{e}_2 = \mathbf{l}$. The unperturbed linear block is $+I_2$, not $-I_2$. Its minimal response is therefore

$$\mathcal{C}_{\perp}^{(\gamma)}(\Omega) = \begin{pmatrix} 1 - \eta_{\gamma} + q_{\gamma} & -\kappa_{\gamma} \\ \kappa_{\gamma} & 1 - \eta_{\gamma} - q_{\gamma} \end{pmatrix}. \quad (61)$$

Here η_{γ} describes a reduction of the positive photon correlation, q_{γ} is the symmetric linear-polarization anisotropy, and κ_{γ} is the antisymmetric rotational component. A nonzero κ_{γ} can imitate a small relative analyzer rotation; changes of η_{γ} , q_{γ} , or of the singular values after kinematic compensation provide stronger evidence for a modified correlation structure. At low frequency one may write

$$\begin{aligned} \eta_{\gamma}(\Omega) &= \eta_{\gamma 0} + \eta_{\gamma 2} \Omega^2 + \dots, \\ q_{\gamma}(\Omega) &= q_{\gamma 0} + q_{\gamma 2} \Omega^2 + \dots, \\ \kappa_{\gamma}(\Omega) &= i\kappa_{\gamma 1} \Omega + i\kappa_{\gamma 3} \Omega^3 + \dots. \end{aligned} \quad (62)$$

This is a phenomenological tensor and frequency structure, not a numerical prediction. A microscopic theory must determine which coefficients are nonzero and their magnitude.

12 Proposed Photon Experiment

12.1 Operational summary

The primary data are single counts and two-photon coincidences, time-tagged relative to the mechanical modulation phase. For each analyzer pair, record $N_{++}, N_{+-}, N_{-+}, N_{--}$ and compute

$$C = \frac{N_{++} + N_{--} - N_{+-} - N_{-+}}{N_{++} + N_{--} + N_{+-} + N_{-+}}. \quad (63)$$

Single counts determine the local polarizations,

$$u_i = \frac{N_{+,i}^{(A)} - N_{-,i}^{(A)}}{N_{+,i}^{(A)} + N_{-,i}^{(A)}}, \quad v_j = \frac{N_{+,j}^{(B)} - N_{-,j}^{(B)}}{N_{+,j}^{(B)} + N_{-,j}^{(B)}}. \quad (64)$$

Complete tomography uses the H/V , D/A , and R/L bases. A first proof-of-principle test may use only $C_{11}, C_{22}, C_{12}, C_{21}$, from which

$$\eta_{\gamma} = 1 - \frac{C_{11} + C_{22}}{2}, \quad q_{\gamma} = \frac{C_{11} - C_{22}}{2}, \quad \kappa_{\gamma} = \frac{C_{21} - C_{12}}{2}. \quad (65)$$

12.2 Optical arrangement and modulation

An SPDC source prepares the positively correlated Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|H\rangle_A |H\rangle_B + |V\rangle_A |V\rangle_B), \quad (66)$$

for which $E_0^{(\gamma)} = +\cos 2(\theta_A - \theta_B)$. In phase variables $a = 2\theta_A$ and $b = 2\theta_B$, this is precisely $M_{AB}^{(\gamma)} = +\cos(a - b)$ obtained from the complete phase interval.

The two photons are analyzed in modules A and B that share the same controlled non-inertial motion. Wherever practicable, both modules are mounted on one rigid moving platform. An equivalent implementation uses two matched stages driven synchronously and monitored independently. For harmonic rotation,

$$\phi_A(t) \simeq \phi_B(t) = \phi_0 \cos \Omega t. \quad (67)$$

It is useful to separate the common and differential coordinates,

$$\phi_+(t) = \frac{\phi_A(t) + \phi_B(t)}{2}, \quad \phi_-(t) = \phi_A(t) - \phi_B(t). \quad (68)$$

The target common-frame experiment has $\phi_-(t) \simeq 0$. For the common coordinate $\phi_+(t) = \phi_0 \cos \Omega t$,

$$\begin{aligned} \dot{\phi}_+ &= -\Omega \phi_0 \sin \Omega t, \\ \ddot{\phi}_+ &= -\Omega^2 \phi_0 \cos \Omega t, \\ \dddot{\phi}_+ &= \Omega^3 \phi_0 \sin \Omega t. \end{aligned} \quad (69)$$

Measurements at several Ω and ϕ_0 separate different frequency scalings. Phenomenologically,

$$\delta \mathcal{C}_{ij}(\Omega) = [\chi_{ij}^{(\phi,+)} - i\Omega \chi_{ij}^{(\dot{\phi},+)} - \Omega^2 \chi_{ij}^{(\ddot{\phi},+)} + i\Omega^3 \chi_{ij}^{(\ddot{\phi},+)}] \phi_+(\Omega) + \chi_{ij}^{(\phi,-)}(\Omega) \phi_-(\Omega). \quad (70)$$

The term proportional to ϕ_- measures imperfect synchronization and the ordinary differential basis response. It must be separated from the common response. Indeed, the standard photon law is unchanged by an equal rotation of the two analyzer axes,

$$\cos 2[(\theta_A + \phi_+) - (\theta_B + \phi_+)] = \cos 2(\theta_A - \theta_B). \quad (71)$$

Thus common rotation supplies a particularly clean null geometry: after calibration, any residual dependence on $\dot{\phi}_+$, $\ddot{\phi}_+$, or higher non-inertial variables cannot be attributed to a change of the relative analyzer angle.

Rotation is retained as a calibrated reference because it generates a known change of orientation together with angular velocity and angular acceleration. It is complemented by a translational vibration mode in which the orientation of both analyzers is kept fixed while modules A and B are mounted on a common multiaxial stage or on two synchronized two- or three-axis piezoelectric stages. A computer generates the same reproducible seeded, band-limited pseudorandom drive signals $u_k(t)$ for the two stages. Equal drive voltages alone do not establish equal motion because the stages, mountings, and payloads can have different transfer functions. Therefore a calibrated accelerometer is attached to each module and records

$$\begin{aligned} a_{A,k}(\omega) &= H_{A,k}(\omega) u_k(\omega), \\ a_{B,k}(\omega) &= H_{B,k}(\omega) u_k(\omega), \end{aligned} \quad (72)$$

where $H_{A,k}$ and $H_{B,k}$ are the measured electromechanical transfer functions. Feedback or digital precompensation is used to make the measured accelerations coincide. Define

$$\mathbf{a}_+(t) = \frac{\mathbf{a}_A(t) + \mathbf{a}_B(t)}{2}, \quad \mathbf{a}_-(t) = \mathbf{a}_A(t) - \mathbf{a}_B(t). \quad (73)$$

The common-frame condition is $\mathbf{a}_-(t) \simeq 0$. A piezoelectric actuator cannot sustain uniform acceleration over an unlimited time because its displacement is bounded; it instead supplies zero-mean harmonic or stochastic acceleration over a controlled frequency band.

The independently driven two-stage arrangement is more general than a single moving platform because it permits the correlation of the motion in the two arms to be varied in a controlled way. For each driven component one may introduce the normalized cross-correlation

$$\rho_k(\omega) = \frac{\text{Re } S_{a_{A,k}a_{B,k}}(\omega)}{\sqrt{S_{a_{A,k}a_{A,k}}(\omega)S_{a_{B,k}a_{B,k}}(\omega)}}, \quad -1 \leq \rho_k \leq 1. \quad (74)$$

Thus $\rho_k \simeq 1$ realizes common motion, $\rho_k \simeq 0$ realizes independent motion, and $\rho_k \simeq -1$ realizes an oppositely driven differential control. A single rigid platform carrying both analysis modules is the experimentally simpler common-frame limit, for which $\mathbf{a}_- \simeq 0$ and $\phi_- \simeq 0$. The two-stage configuration shown in Fig. 1 is retained because it allows continuous interpolation between common, partially correlated, independent, and differential motion.

The linear translational response is written

$$\delta\mathcal{C}_{ij}(\omega) = \sum_{k=x,y,z} \left[\chi_{ijk}^{(a,+)}(\omega) a_{+,k}(\omega) + \chi_{ijk}^{(a,-)}(\omega) a_{-,k}(\omega) \right], \quad (75)$$

where $\chi^{(a,+)}$ is the sought susceptibility to the common non-inertial motion and $\chi^{(a,-)}$ quantifies differential motion and local artifacts. Different common drive directions probe different components of the non-inertial susceptibility. With $\mathbf{z} = (\mathbf{a}_+, \mathbf{a}_-)^T$, both susceptibilities are estimated from the correlation–acceleration cross-spectral matrix,

$$(\chi_{ij}^{(a,+)}, \chi_{ij}^{(a,-)}) = \mathbf{S}_{C_{ijz}}(\omega) \mathbf{S}_{zz}^{-1}(\omega). \quad (76)$$

The seeded waveform makes repeated runs and null tests directly comparable, whereas the two accelerometers supply the physical common and differential inputs used in the analysis.

12.3 Phase-synchronous reconstruction and controls

Each event is assigned the phase

$$\varphi_\Omega = \Omega t \pmod{2\pi}. \quad (77)$$

For M phase bins, the first harmonic estimator is

$$\tilde{\mathcal{C}}_{ij}^{(1)} = \frac{2}{M} \sum_{m=1}^M \mathcal{C}_{ij}(\varphi_m) e^{-i\varphi_m}. \quad (78)$$

Four modes are compared: (I) stationary baseline; (II) stationary synchronous optical modulation of both analyzers reproducing the same basis transformations; (III) common harmonic rotation with $\phi_A(t) \simeq \phi_B(t)$; and (IV) common seeded multiaxial piezoelectric vibration with $\mathbf{a}_A(t) \simeq \mathbf{a}_B(t)$. A deliberately imposed differential drive provides an additional calibration of local artifacts. The optical–mechanical comparison is supplemented by two-sided kinematic compensation,

$$\tilde{\mathcal{C}}(t) = R_A^{-1}(t) \mathcal{C}_{\text{meas}}(t) R_B^{-T}(t), \quad \Delta\mathcal{C}_{\text{res}}(t) = \tilde{\mathcal{C}}(t) - C^{(0)}. \quad (79)$$

Reversing the rotation direction tests

$$\eta_\gamma(-\Omega) = \eta_\gamma(\Omega), \quad q_\gamma(-\Omega) = q_\gamma(\Omega), \quad \kappa_\gamma(-\Omega) = -\kappa_\gamma(\Omega). \quad (80)$$

In Mode IV the principal observables are the cross spectra with both $\mathbf{a}_+$ and $\mathbf{a}_-$, their full autospectral matrix, and the corresponding magnitude-squared and partial coherences. A common-frame candidate must follow $\mathbf{a}_+$ after conditioning on $\mathbf{a}_-$; a response that follows only $\mathbf{a}_-$ is classified as differential motion or a local artifact. Repeating the same seeded waveform with the optical beam blocked, with separable photons, and with a mechanically isolated dummy payload identifies electronic, detector, and mounting artifacts.

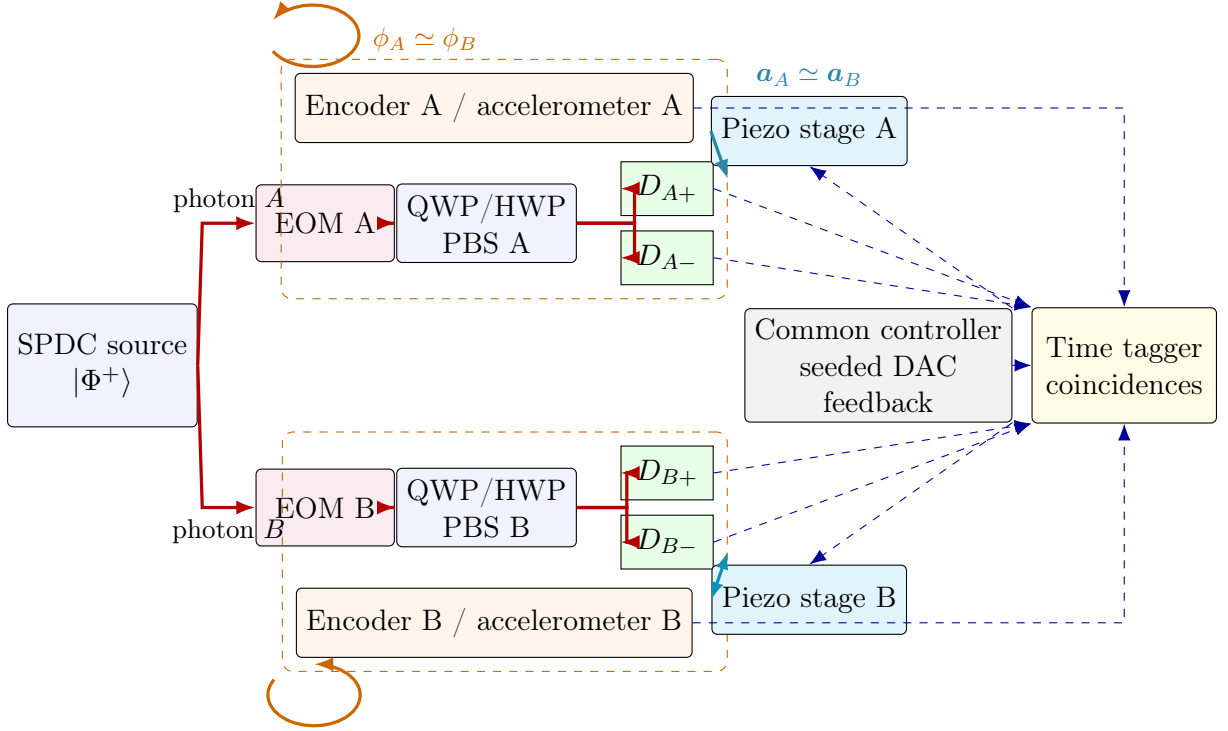


Figure 1: Two-arm photon experiment with controllable motion correlation. The two analysis modules can undergo synchronous, partially correlated, independent, or differential motion. Mode II applies equivalent synchronous optical control with EOM A and EOM B. In Mode III both modules rotate with $\phi_A(t) \simeq \phi_B(t)$; in Mode IV two feedback-controlled piezoelectric stages apply the same seeded multiaxial vibration, so that $\mathbf{a}_A(t) \simeq \mathbf{a}_B(t)$. Independent encoders and accelerometers measure the common variables ϕ_+ and $\mathbf{a}_+$ and the residual differential variables ϕ_- and $\mathbf{a}_-$. A single rigid moving platform carrying both modules gives the experimentally simpler common-frame limit, whereas the two-stage arrangement permits the motion correlation between the arms to be varied deliberately. Solid red lines denote optical paths and dashed blue lines synchronization and measurement signals.

For N_{ij} detected pairs, a leading statistical estimate is

$$\sigma(C_{ij}) \simeq \sqrt{\frac{1 - C_{ij}^2}{N_{ij}}}, \quad (81)$$

while a full analysis should propagate the Poisson uncertainties of the four coincidence counts, including accidental coincidences and dark counts. The minimum resolvable susceptibility scales as

$$\chi_{ij}^{\min} \sim \frac{\sigma(\tilde{C}_{ij}^{(1)})}{\xi_0}. \quad (82)$$

Principal systematic effects include analyzer-angle error, unintended angular motion, optical-path modulation, detector-efficiency modulation, source drift, accidental coincidences, decoherence, and thermal drift. Single counts, two independent encoders, two three-axis accelerometers, beam-position monitoring, shifted coincidence windows, interleaved control runs, and repeated state tomography are therefore required. For an ideal compensated $|\Phi^+\rangle$ state, the standard null prediction is

$$\tilde{\mathcal{C}}^{(\gamma)}(t) = \text{diag}(1, 1, -1), \quad \eta_\gamma(\Omega) = q_\gamma(\Omega) = \kappa_\gamma(\Omega) = 0. \quad (83)$$

13 Discussion

The distinction $E_\gamma = +\cos\theta$ versus $E_f = -\cos\theta$ concerns the sign and geometry of the photon and fermion correlation sectors. It does not alter Bell's theorem and does not assume that non-inertial motion necessarily changes quantum nonlocality. Its purpose is to distinguish a change of scalar projections from a change of the reconstructed correlation object. A variation of E or S alone may result entirely from a rotation of local bases. Singular values of the compensated correlation matrix and an independently reconstructed density matrix provide stronger tests.

Three outcomes are possible: a purely kinematic response; a residual fully accounted for by standard optical or relativistic transformations; or an additional reproducible residual. Only the third requires an extended dynamical interpretation. A candidate signal must be phase locked or spectrally coherent with the measured motion, reproducible in independent runs, scale consistently with ϕ_0 , Ω , and the acceleration spectrum, transform correctly under rotation reversal and changes of vibration direction, be coherent with the common variables ϕ_+ or $\mathbf{a}_+$ but not explainable by the measured differential variables ϕ_- or $\mathbf{a}_-$, and remain after all calibrated optical, mechanical, and relativistic backgrounds are included. Common motion of both analysis arms is therefore the primary test of a shared non-inertial variable. Single-arm or deliberately mismatched motion is retained as a differential control, not interpreted as the sought common-frame response. Standard quantum theory predicts that an exactly common calibrated rotation cancels from the relative analyzer angle and that the compensated tensor remains unchanged; a nonzero common residual is consequently a falsifiable hypothesis rather than an assumed effect.

14 Conclusion

The photon correlation obtained by averaging two projection amplitudes over the complete phase interval $0 \leq \lambda < 2\pi$ is $M_{AB}^{(\gamma)}(a, b) = +\cos(a - b)$. For physical linear-polarizer axes this becomes $E_\gamma = +\cos 2(\theta_A - \theta_B)$. The fermionic singlet remains $E_f = -\cos(\alpha_A - \alpha_B)$ because the complete phase-momentum exchange loop carries the negative holonomy. The full phase interval and the projective periodicity of linear polarization are therefore distinct geometric elements and must not be conflated.

The associated unbiased binary probabilities reproduce these signed cosine laws and lead to opposite signed optimal CHSH combinations with the same absolute Tsirelson value. The non-inertial construction therefore remains inside the quantum correlation bound while retaining an operational distinction between the two exchange sectors.

A Bell observable is a scalar projection of a complete two-particle correlation tensor. For the photon state $|\Phi^+\rangle$ the baseline Stokes block is $C^{(\gamma,0)} = \text{diag}(1, 1, -1)$, whereas for the fermionic singlet $C^{(f,0)} = -I$. For a modulated system we introduced the extended tensor $\mathcal{T}_{\mu\nu}(\boldsymbol{\xi})$ and its susceptibility $X_{\mu\nu\alpha}(\Omega)$. Phase-synchronous Stokes tomography, comparison with equivalent optical modulation, common harmonic rotation and matched multiaxial piezoelectric vibration of both arms, explicit common-differential decomposition, and two-sided kinematic compensation define complementary null tests for an additional non-inertial response. The construction replaces isolated scalar variations by a reconstructible dynamical tensor and provides an experimentally testable framework for Bell correlations in non-inertial reference frames.

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